Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller

Interface Design Description

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1.0 Purpose and Scope

This document defines each interface identified in the architecture [2, Sec. 7.0]: the format and content of every data element that crosses it, its communication method and protocol, and the signals between the two units. It covers the minimum content of an interface design description [1, Sec. 1].

2.0 Documents

[1] NASA, *NASA Software Engineering Handbook*, NASA-HDBK-2203, Version D, Topic 5.02 “IDD – Interface Design Description”.
[2] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Software Architecture*, Issue 1, Revision 0, Draft.
[3] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Software Requirements Specification*, Issue 1, Revision 0, Draft.
[4] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Hazard Analysis*, Issue 1, Revision 1, Draft.
[5] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Concept of Operations*, Issue 1, Revision 1, Draft.
[6] S. Josefsson and I. Liusvaara, *Edwards-Curve Digital Signature Algorithm (EdDSA)*, RFC 8032, January 2017.
[7] NIST, *Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC*, SP 800-38D, November 2007.
[8] Arm, *ARM Debug Interface v5 Architecture Specification*, IHI 0031A, 2006.
[9] STMicroelectronics, *STM32G4 Series Reference Manual*, RM0440 Rev 9.
[10] STMicroelectronics, *How to Use the USART Protocol in the Bootloader on STM32 MCUs*, AN3155 Rev 22.
[11] STMicroelectronics, *Introduction to System Memory Boot Mode on STM32 MCUs*, AN2606 Rev 70.
[12] Texas Instruments, *RS-422 and RS-485 Standards Overview and System Configurations*, SLLA070D, May 2010.
[13] Texas Instruments, *Introduction to the Controller Area Network (CAN)*, SLOA101B, May 2016.

[14] MCUboot project, *Bootloader* design document, `docs/design.md`, commit 1d2643c, 11 September 2026.

3.0 Overview

Figure 1 shows every interface and the signals it uses. Table 1 gives each interface's type, priority and use, and whether it is safety-critical [1, Sec. 1].

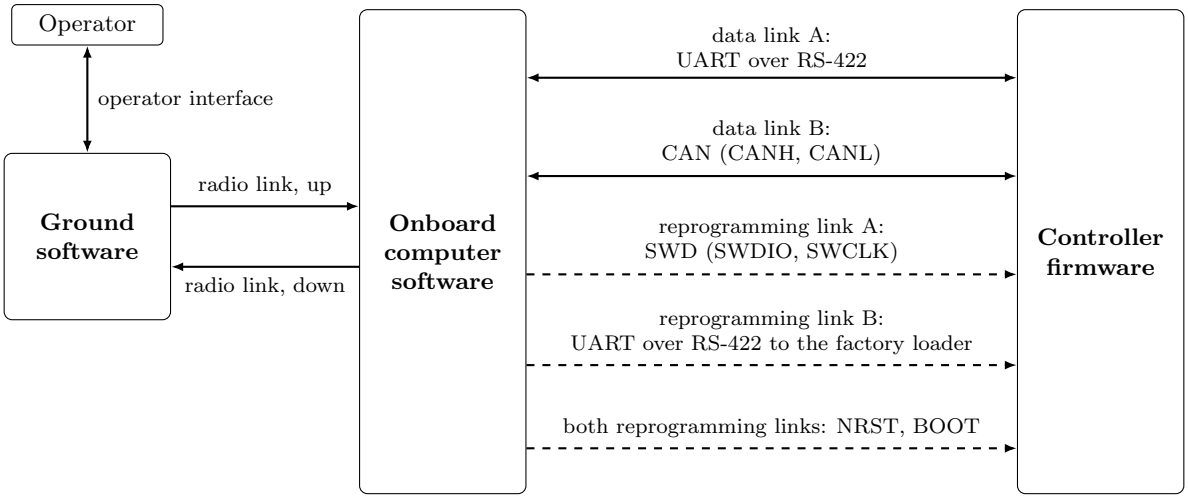


Figure 1: The interfaces and their signals. Solid: used in operation. Dashed: used only to restore or reload the controller.

Interface	Type	Priority	Used	Safety-critical
Operator interface	Commands in, state out	ONLY	ALWAYS	YES
Radio link	Data transfer, encrypted and signed	ONLY	AT CONTACTS	YES
Data link A	Data transfer	PRIMARY	ALWAYS	YES
Data link B	Data transfer	BACKUP	ALWAYS	YES
Reprogramming link A	Memory load	PRIMARY	RESTORE	YES
Reprogramming link B	Memory load	BACKUP	RESTORE	YES

Table 1: Interface types.

Legend. Priority: ONLY, the one interface for its traffic; PRIMARY, used first; BACKUP, used when the primary one is lost. Used: ALWAYS, whenever the System runs; AT CONTACTS, during contact windows; RESTORE, during a restore or a reload through the reprogramming link.

4.0 Message Layer

Every command, answer and report travels in the same frame on the radio link and on both data links (Figure 2). Multi-byte fields are little-endian. The check value is a CRC-32 of every byte before it. The receiver computes it again; a different result means the frame was damaged on the way. The polynomial is 0x4C11DB7, the one the controller's processor computes in hardware [9, Sec. 16.2, p. 454]. The calculation starts from 0xFFFFFFFF, takes each byte least significant bit first, and inverts the result. A receiver drops a frame whose check value or version it does not accept.

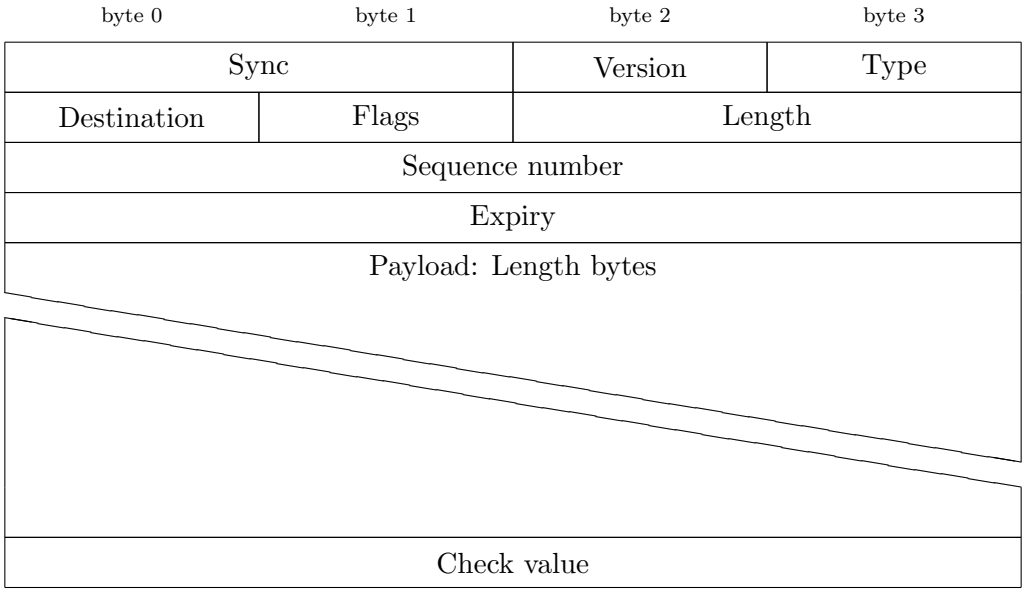


Figure 2: The frame. Each row is four bytes. The header is the first 16 bytes.

Field	Bytes	Values	Meaning
Sync	2	0xA5, 0x5A	Start of a frame
Version	1	0x01	Frame format version
Type	1	0x01 command, 0x02 answer, 0x03 report	What the payload holds
Destination	1	0x01 onboard computer, 0x02 controller, 0x03 ground	Who acts on the frame
Flags	1	bit 0	Set when the onboard computer releases a scheduled command; other bits 0
Length	2	0 to 1024	Payload length in bytes
Sequence number	4	1 to $2^{32} - 1$	One higher than the sender's last frame [3, REQ-065]
Expiry	4	seconds of ground time	After it, the command is refused [3, REQ-066, REQ-133]; 0 in answers and reports
Payload	Length		The command, answer or report
Check value	4	CRC-32	Detects damage to the frame [3, REQ-107]

4.1 Radio Envelope

On the radio link each frame travels inside an envelope (Figure 3). The frame is signed twice with Ed25519, an elliptic-curve digital signature scheme, each signature 64 bytes [6, Sec. 1]. The frame and both signatures are then encrypted with AES-256 in Galois/Counter Mode, with a 12-byte nonce and a 16-byte tag [7, Secs. 5.2.1.1–5.2.1.2, pp. 8–9]. The link encryptor builds the envelope. The onboard computer opens it, checks both signatures, and acts only when all three checks pass [3, REQ-127 to REQ-129].

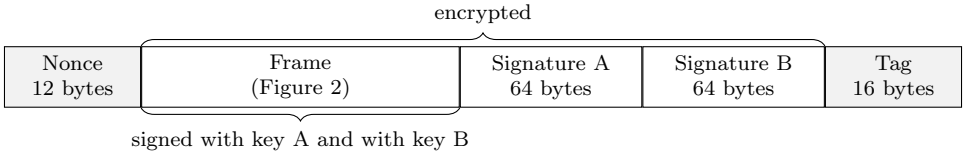


Figure 3: The radio envelope. The tag proves the encrypted part unchanged and encrypted with the right key; the signatures prove the frame comes from the operator.

4.2 Answers

Every command gets two answers [3, REQ-033]: RECEIVED when it passes the checks on arrival, then DONE with its result or REFUSED with a reason (Figure 4). A command sent again with a sequence number already run is answered ALREADY DONE and is not run twice [3, REQ-031]. An envelope that fails any of its checks is dropped and logged without an answer, so that a sender without the keys learns nothing [3, REQ-064].

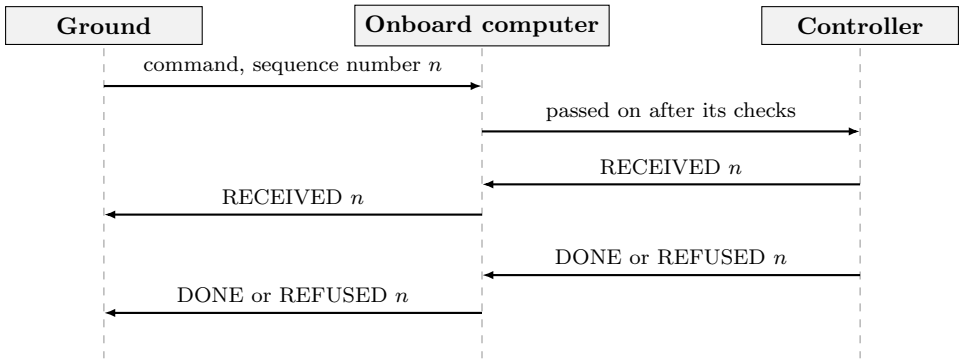


Figure 4: A command to the controller and its two answers. DONE carries the result, REFUSED the reason. A command to the onboard computer is answered by the onboard computer.

An answer's payload is six bytes: the sequence number answered (4 bytes); the kind (1 byte): 0x01 RECEIVED, 0x02 DONE, 0x03 REFUSED, 0x04 ALREADY DONE; and a code (1 byte): for DONE, 0x00 on success; for REFUSED, the reason below. The ground and the onboard software share this list [3, REQ-100].

Code	Reason	Requirement
0x01	Dangerous command without a valid arming command	REQ-029, REQ-112
0x02	Arming command expired	REQ-030
0x03	Command past its expiry	REQ-066, REQ-133
0x04	Old command sent again	REQ-065
0x05	Not allowed in the current mode	REQ-027
0x06	Impossible values, or an unknown command	REQ-028
0x07	No new firmware during a trial	REQ-050
0x08	Image damaged	REQ-005, REQ-044
0x09	Image not from the operator	REQ-006, REQ-043
0x0A	Image older than the minimum version	REQ-045
0x0B	Image built for another controller	REQ-046
0x0C	The controller's clock disagrees with the scheduled time	REQ-134
0x0D	A dangerous command cannot be scheduled	REQ-029, REQ-112
0x0E	Image built for the other slot	REQ-005
0x0F	Trial tests not passed	REQ-012
0x10	Image of the wrong kind for its place	REQ-005, REQ-008
0x11	Boot state could not be stored	REQ-012, REQ-021

5.0 Commands, Reports and Firmware Images

5.1 Commands

A command's payload is its one-byte identifier followed by its fields. Identifiers 0x01 to 0x1F are commands to the onboard computer, 0x20 to 0x3F commands to the controller. UPLOAD BEGIN to UPLOAD CANCEL also go to the controller, with Destination 0x02, but only from the onboard computer, as the steps of a DELIVER. The onboard computer refuses them from the ground with 0x06, so every image reaches the controller after the onboard computer's own checks [3, REQ-047]. The controller writes the image into its slot that is not confirmed, so it can never write over the firmware it runs [3, REQ-007]. During a trial, the controller accepts them only for the test image of its trial tests, image number 0xFFFF. The test image's last four bytes are the CRC-32 of the bytes before them. The controller keeps the test image in working memory, checks its check value, and writes nothing into either slot [3, REQ-139].

ID	Command	Fields (bytes)	Modes	Dangerous	Requirement
0x01	UPLOAD BEGIN	image number (2), image length (4)	ANY; controller: OPERATE, FALLBACK; TRIAL, test image only	NO	REQ-049
0x02	UPLOAD CHUNK	image number (2), offset (4), data (up to 256)	ANY; controller: OPERATE, FALLBACK; TRIAL, test image only	NO	REQ-049
0x03	UPLOAD END	image number (2)	ANY; controller: OPERATE, FALLBACK; TRIAL, test image only	NO	REQ-043 to 047
0x04	UPLOAD CANCEL	image number (2)	ANY; controller: OPERATE, FALLBACK; TRIAL, test image only	NO	REQ-051
0x05	DELIVER	image number (2)	OPERATE, FALLBACK	NO	REQ-048, REQ-050
0x06	SET LIMITS	minimum version (4), controller identity (2)	ANY	YES	REQ-053
0x07	RESTORE	none	ANY	YES	REQ-055
0x08	REPROGRAM	image number (2), region (1): bootloader, main firmware or last-resort firmware	ANY	YES	REQ-058, REQ-121
0x09	SET ONBOARD TIME	ground time (4)	ANY	YES	REQ-073
0x0A	SCHEDULE	run time (4), one complete frame	ANY	NO	REQ-068
0x0B	SEND REPORTS	first report number (4)	ANY	NO	REQ-074
0x0C	REPLACE KEY	which key (1): signing A, signing B or encryption; new key (32)	ANY	YES	REQ-084 to 086
0x0D	ARM	command identifier (1), its sequence number (4)	ANY	NO	REQ-029, REQ-112

ID	Command	Fields (bytes)	Modes	Dangerous	Requirement
0x20	SWITCH	none	OPERATE, FALLBACK	YES	REQ-021
0x21	RETURN TO PREVIOUS	none	OPERATE, TRIAL	YES	REQ-113
0x22	START FAILED	which copy (1)	OPERATE, FALLBACK	YES	REQ-019
0x23	LEAVE FALLBACK	none	FALLBACK	NO	REQ-117
0x24	SET CONTROLLER TIME	ground time (4)	OPERATE, TRIAL, FALLBACK	YES	REQ-132
0x25	GET STATUS	none	OPERATE, TRIAL, FALLBACK	NO	REQ-081
0x26	CONFIRM	none	TRIAL	YES	REQ-012, REQ-020
0x40	EQUIPMENT	set by the controlled equipment	**[TBD-2]**	**[TBD-2]**	—

Legend. Modes: the controller modes in which the command is accepted, within those that accept commands [5, Sec. 3.4]; ANY, every mode, RECOVERY included. Dangerous: YES, it needs an arming command first [3, REQ-029, REQ-112]; NO, it does not.

5.2 Arming

An ARM command names the one dangerous command it allows, by identifier and sequence number. It allows no other command, and it expires after **[TBD-3]** [3, REQ-030]. The onboard computer and the controller each check it [3, REQ-029, REQ-112].

The onboard computer is the only sender on a data link, so a command it passes on carries the onboard computer's sequence number there, not the ground's (Section 4.0). For a dangerous command to the controller, the onboard computer holds the ground's ARM and checks it. When the named command arrives, it sends the two as a pair: an ARM naming the sequence number the command will carry on the data link, then the command, with no frame between them.

5.3 Scheduled Commands

SCHEDULE holds a complete frame on the onboard computer until its run time [3, REQ-068]. At that time the onboard computer checks the frame again against its check value [3, REQ-136] and passes it to the controller with Flags bit 0 set. The controller runs it only if its own clock agrees with the run time within **[TBD-4]** [3, REQ-134]. A dangerous command cannot be scheduled, because no operator is present at run time to arm it.

5.4 Reports

The controller sends a HEALTH report at **[TBD-5]** and an EVENT report as soon as an event happens [3, REQ-037]. The onboard computer adds the report number and its own clock time before storing the report [3, REQ-070, REQ-071, REQ-135].

Report	Field	Bytes	Set by
ALL	report number	4	onboard computer
ALL	onboard computer clock time	4	onboard computer
ALL	controller clock time	4	controller
ALL	report kind: 0x01 HEALTH, 0x02 EVENT	1	controller
HEALTH	mode: 0x01 OPERATE, 0x02 TRIAL, 0x03 FALLBACK	1	controller
HEALTH	running firmware: kind (1), version (4)	5	controller
HEALTH	running slot: 0x01 A, 0x02 B, 0x03 last-resort	1	controller
HEALTH	failed starts of the new copy	1	controller

Report	Field	Bytes	Set by
HEALTH	trial time left, in seconds	2	controller
HEALTH	last self-test: 0x00 PASSED, 0x01 FAILED	1	controller
EVENT	event code	2	controller or onboard computer
EVENT	detail	up to 32	controller or onboard computer

Legend. Report: ALL, every report; HEALTH or EVENT, that kind only.

The event codes:

Code	Event	Sent by	Requirement
0x0001	fault detected	controller	REQ-038
0x0002	fall-back to the previous copy; detail: refused copy (1), check result (1)	controller	REQ-040
0x0003	fall-back to the last-resort firmware; detail: refused copy (1), check result (1)	controller	REQ-040
0x0004	self-test failed	controller	REQ-039
0x0005	change in memory detected	controller	REQ-041
0x0006	trial started	controller	REQ-037
0x0007	firmware confirmed	controller	REQ-037
0x0008	firmware given up	controller	REQ-037
0x0009	restart	controller	REQ-037
0x000A	trial tests passed	controller	REQ-139
0x000B	trial tests failed; detail: which test (1): 0x01 self-test, 0x02 data link, 0x03 test image	controller	REQ-139, REQ-140
0x0101	data link lost	onboard computer	REQ-081
0x0102	data link working again	onboard computer	REQ-082
0x0103	restore done; detail: its result	onboard computer	REQ-060
0x0104	health reports dropped; detail: count and time span	onboard computer	REQ-076
0x0105	onboard computer clock differs from ground time	onboard computer	REQ-072
0x0106	controller restarted by the onboard computer; detail: reason (1): 0x01 trial past its time	onboard computer	REQ-013

In events 0x0002 and 0x0003, the refused copy uses the codes of the HEALTH report's running slot: 0x01 slot A, 0x02 slot B, 0x03 last-resort firmware. The check result names the first check the copy failed, so the operator can tell, for example, which of the two signatures failed:

Check result	Check that failed
0x00	none: the copy passed
0x01	the copy is empty
0x02	proprietary string
0x03	format
0x04	kind
0x05	target
0x06	length
0x07	start address outside the region
0x08	body check value
0x09	signature A
0x0A	signature B

5.5 Firmware Images

Every firmware image, of every kind, has the same layout (Figures 5 and 6), so that the restore checks the bootloader and the last-resort firmware with the same checks as an update [3, REQ-118]. Both signatures cover the header as well as the body, so that the version and target in the header cannot be changed without breaking them; MCUboot's signature covers the same two parts [14, Sec. Image format]. The controller checks both before it runs the image [3, REQ-005, REQ-006]. Images move in chunks of 256 bytes, the most the factory loader writes at once [10, p. 7], so the onboard computer can write a chunk through the loader as it received it.

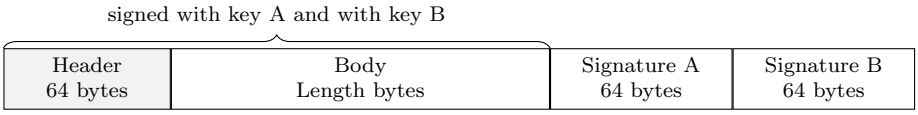


Figure 5: A firmware image.

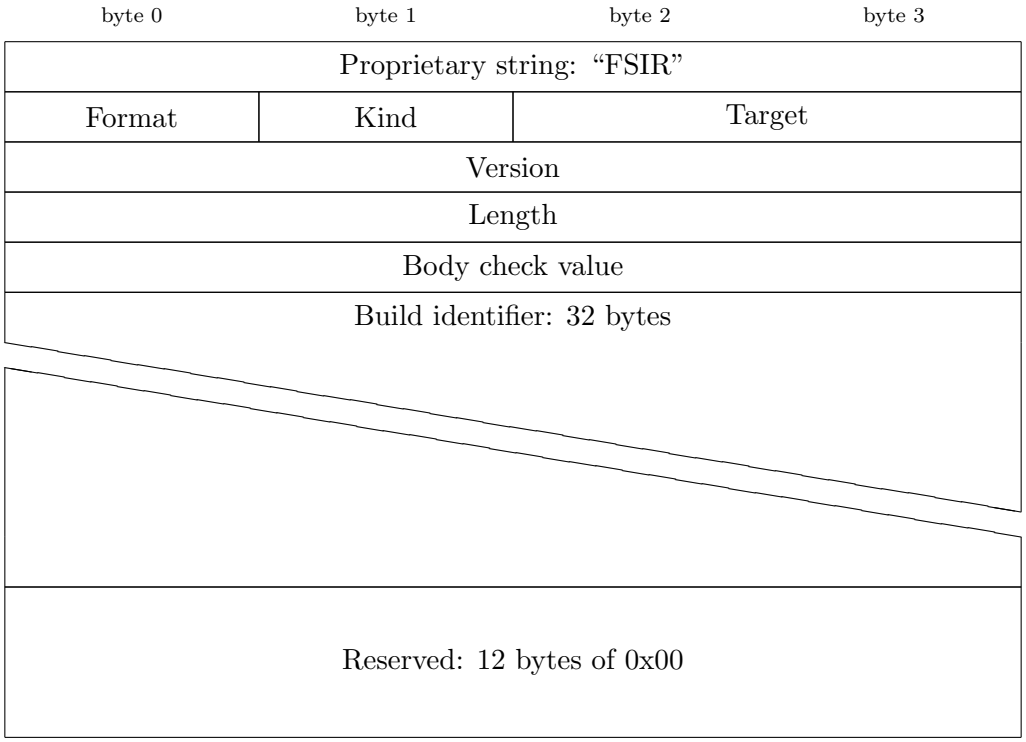


Figure 6: The image header. Proprietary string: four fixed bytes, 0x46 0x53 0x49 0x52, that mark the start of an image; the checker rejects any other value at once. MCUboot's image header also starts with a fixed 4-byte value [14, Sec. Image format]. Kind: 0x01 bootloader, 0x02 main firmware, 0x03 last-resort firmware. Target: the controller identity [3, REQ-046]. Body check value: CRC-32 of the body, computed as for the frame. Build identifier: names the build that was tested [3, REQ-090].

An update uses the commands in this order: UPLOAD BEGIN, UPLOAD CHUNK and UPLOAD END to bring the image to the onboard computer (steps 2–3 of [2, Fig. 2]); DELIVER, after which the onboard computer sends the same UPLOAD commands to the controller (step 4); then ARM and SWITCH to start it (step 6). During the trial the onboard computer sends the test image, the controller runs its trial tests and reports the result, and the operator keeps the new firmware with ARM and CONFIRM (step 7). Each command gets its two answers (Section 4.2): the answers to UPLOAD CHUNK tell the ground where to resume a broken upload [3, REQ-049], and the answer to UPLOAD END says whether the image passed its checks. The trial's progress reaches the ground as event reports: trial started; trial tests passed or failed; firmware confirmed or given up (Section 5.4). Figure 7 shows the whole exchange.

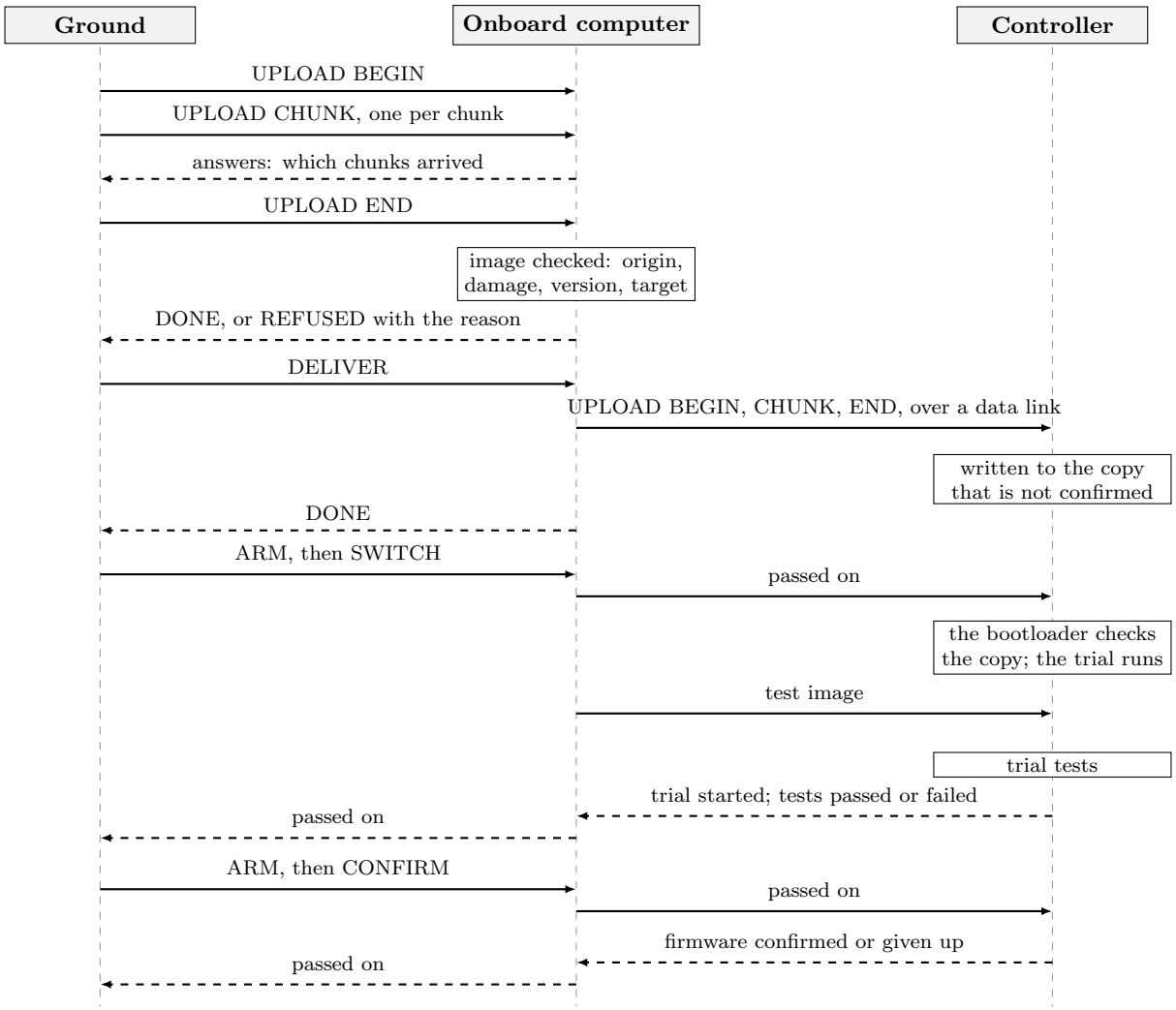


Figure 7: An update. Solid: commands and the image. Dashed: answers and reports. Messages to and from the ground travel only during contacts. If a trial test fails, the controller returns to its previous copy at once and no CONFIRM follows.

6.0 Links

The controller sits in a separate unit from the onboard computer, reached through a cable harness. The links between the two units use balanced line drivers, made for long distances and noisy environments [12, Sec. 1, p. 3].

6.1 Radio Link

The radio equipment lies outside the System and is used as given [5, Sec. 1.2.2]. The ground software and the onboard computer software exchange envelopes (Section 4.1) through it.

6.2 Data Link A: UART over RS-422

The UART runs at 115 200 baud, 8 data bits, no parity, 1 stop bit; a frame with a gap of more than 10 ms inside it is dropped. Each direction has its own RS-422 line, because an RS-422 line has one driver [12, Sec. 2, p. 3]. RS-422 sets no maximum cable length [12, Sec. 2.1, p. 3]. On each data link the onboard computer sends GET STATUS every **[TBD-6]**. A link that misses

[TBD-7] answers in a row is reported lost [3, REQ-081] and is used again once it answers [3, REQ-082].

6.3 Data Link B: CAN

Data link B uses CAN FD frames, which carry up to 64 data bytes [9, Sec. 44.2, p. 1953], at a bit rate of **[TBD-8]**. Either end sends whenever it has a frame [13, Sec. 3, p. 3]. A CAN frame received with an error is sent again [13, Sec. 3.1, p. 3]. A longer frame is split into segments (Figure 8): each CAN frame carries a segment number in its first byte, then up to 63 bytes of the frame. Segment 0 starts a frame; the receiver joins the segments in order and checks the whole frame with its check value. Each direction uses its own CAN identifier.

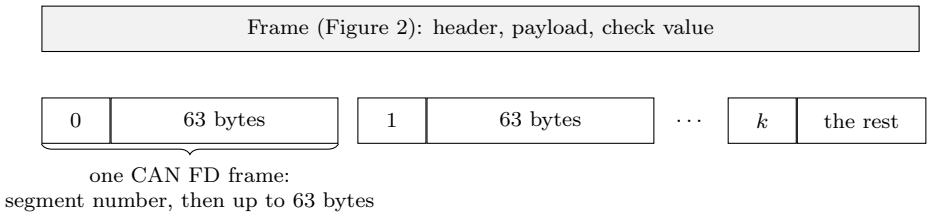


Figure 8: A frame split into CAN FD frames.

6.4 Reprogramming Link A: SWD

The onboard computer drives SWD from two general-purpose pins, with the controller held stopped, so that no controller software runs (Figure 9).

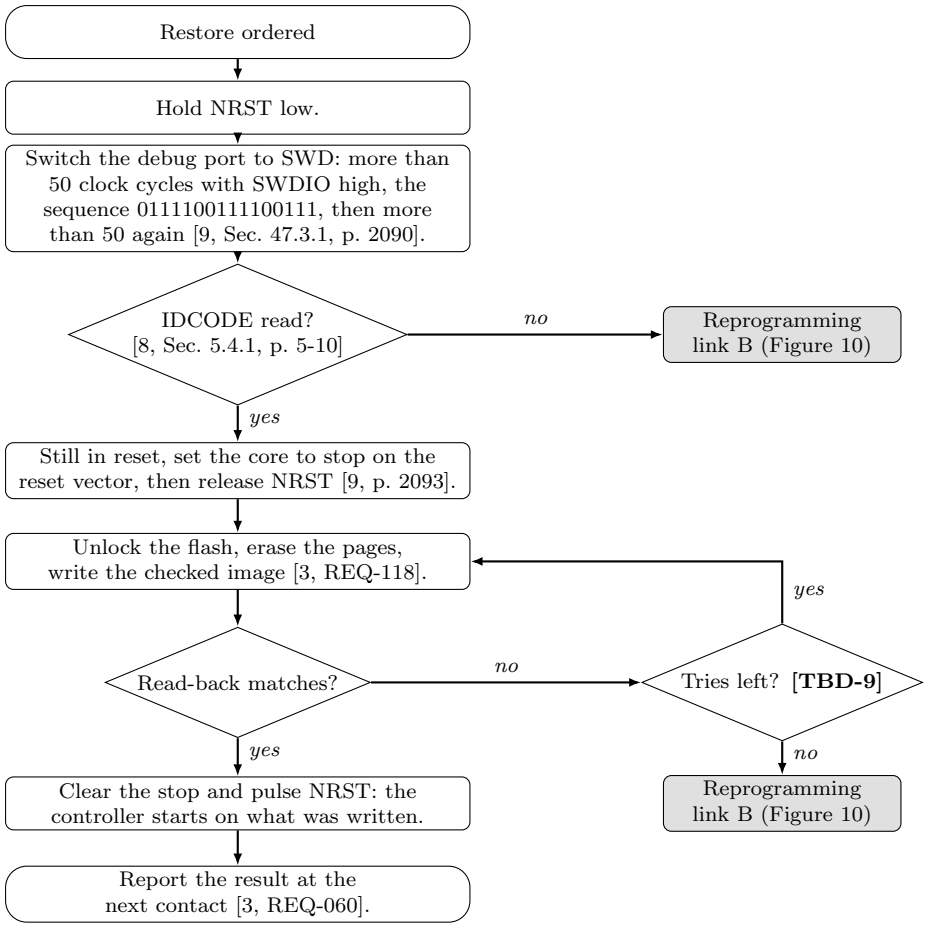


Figure 9: A restore through SWD. Each diamond is a check with a yes and a no exit.

6.5 Reprogramming Link B: Factory Loader

The factory loader's UART runs over its own RS-422 lines, as data link A does. With the option bits as shipped (BOOT_LOCK = 0, nBOOT1 = 1, nSWBOOT0 = 1), the controller starts its factory loader when BOOT is high at reset [11, Table 2, p. 36]. The loader's protocol is fixed by its maker (Figure 10): 8 data bits, even parity, 1 stop bit [11, Table 115, p. 274]; the host sends 0x7F and the loader answers ACK, 0x79 [10, p. 5]; every command byte is followed by its complement, every block ends with the XOR of its bytes, and each packet is answered ACK 0x79 or NACK 0x1F [10, p. 8].

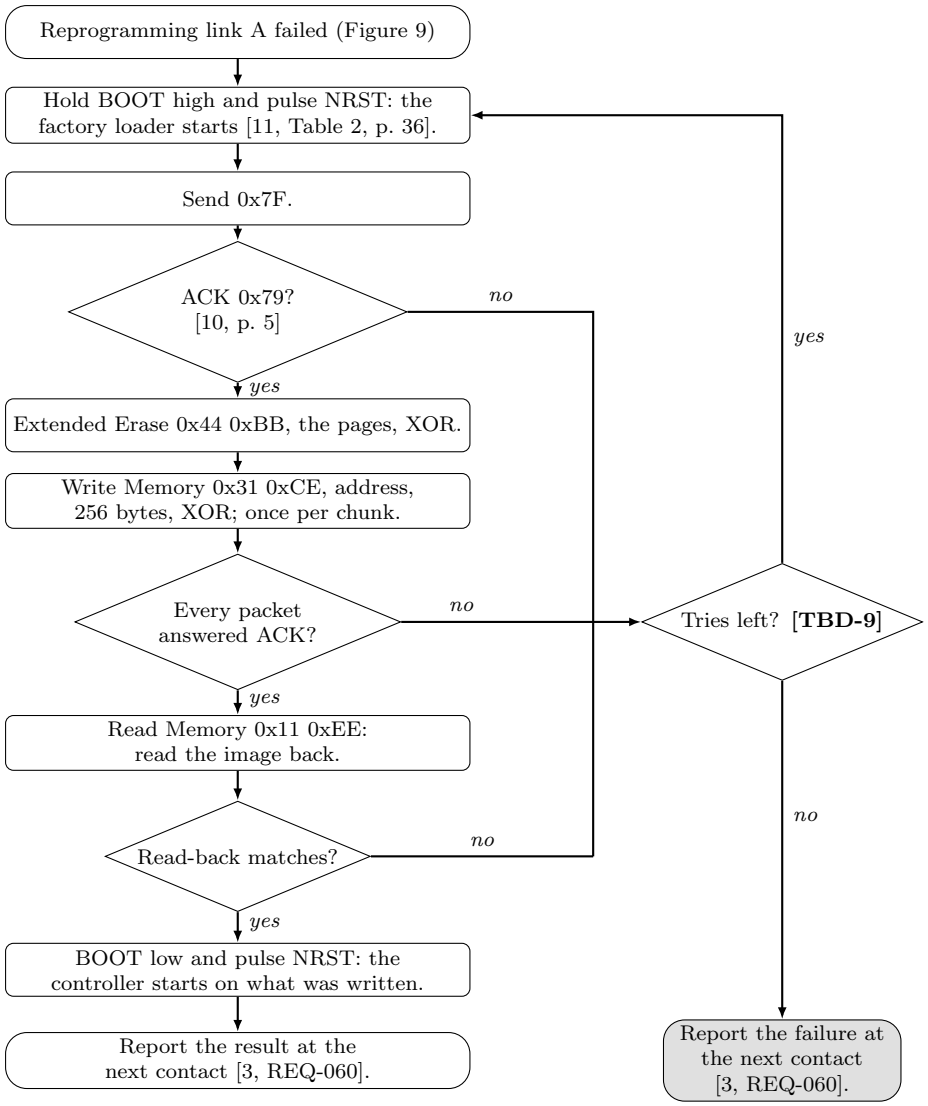


Figure 10: A restore through the factory loader. Each diamond is a check with a yes and a no exit. Command codes from [10, p. 7].

The onboard computer never sends the loader's Readout Protect (0x82) or Write Protect (0x63) commands [10, p. 7]: either would lock a way back in [3, REQ-125].

6.6 Operator Interface

The operator console shows every command with its state (sent, RECEIVED, DONE or REFUSED with the reason), every command still waiting for an answer [3, REQ-094], which firmware runs [3, REQ-095], and every report received. It accepts commands, planned sequences of commands [3, REQ-115] and firmware releases. Signing with key B needs the operator's hardware security key [2, Sec. 5.4].

7.0 Signals Between the Units

The signals that cross the cable harness between the onboard computer and the controller.

Signal	Interface	Direction	Electrical form
UART line	data link A	TO CONTROLLER	RS-422
UART line	data link A	TO ONBOARD COMPUTER	RS-422
CANH, CANL	data link B	BOTH	CAN
SWDIO	reprogramming link A	BOTH	TBD
SWCLK	reprogramming link A	TO CONTROLLER	TBD
UART line	reprogramming link B	TO CONTROLLER	RS-422
UART line	reprogramming link B	TO ONBOARD COMPUTER	RS-422
NRST	reprogramming links A and B	TO CONTROLLER	TBD
BOOT	reprogramming link B	TO CONTROLLER	TBD

Legend. Direction: TO CONTROLLER, TO ONBOARD COMPUTER, or BOTH. Electrical form: RS-422, one balanced line; CAN, the CAN bus pair; TBD, not yet known [**TBD-10**].

8.0 Safety Features

Each feature of the interfaces that guards against a hazard of the hazard analysis [4, Sec. 4.0].

Feature	Interfaces	Hazards	Requirements
Check value on every frame and image body	radio link, data links	H-7, H-9	REQ-044, REQ-107
Encryption and two signatures	radio link	H-2, H-5	REQ-127 to 129
Sequence numbers	radio link, data links	H-6	REQ-031, REQ-065
Expiry in every command	radio link, data links	H-6	REQ-066, REQ-133
Arming before a dangerous command	radio link, data links	H-6	REQ-029, REQ-112
Two answers to every command	radio link, data links	H-11	REQ-033
Failed envelopes dropped and logged	radio link	H-5	REQ-064
Link check on each data link	data links	H-4	REQ-081
CAN check value, and resend of a damaged CAN frame	data link B	H-7, H-9	REQ-107
Only checked images written	reprogramming links	H-2, H-7	REQ-118
No protection command sent to the loader	reprogramming link B	H-1	REQ-125

A Acronyms

ACK Acknowledge
AES Advanced Encryption Standard
CAN Controller Area Network
CANH CAN high line
CANL CAN low line
CRC Cyclic redundancy check
GCM Galois/Counter Mode
NACK Not acknowledge
NRST Reset pin
SWCLK SWD clock line
SWD Serial Wire Debug
SWDIO SWD data line
TBD To Be Determined
TBR To Be Resolved
UART Universal asynchronous receiver-transmitter

B Traceability

Each requirement on an interface, and the section that meets it.

Requirement	Section
REQ-064 to 066, REQ-107	4.0 Message Layer
REQ-127 to 129	4.1 Radio Envelope
REQ-027, REQ-028, REQ-031, REQ-033 to 035, REQ-100	4.2 Answers
REQ-019, REQ-021, REQ-053, REQ-055, REQ-058, REQ-073, REQ-074, REQ-084 to 086, REQ-113, REQ-117, REQ-121, REQ-132	5.1 Commands
REQ-029, REQ-030, REQ-112	5.2 Arming
REQ-068, REQ-069, REQ-134, REQ-136	5.3 Scheduled Commands
REQ-037 to 041, REQ-060, REQ-070 to 072, REQ-076, REQ-135	5.4 Reports
REQ-005, REQ-006, REQ-043 to 051, REQ-090	5.5 Firmware Images
REQ-080 to 082, REQ-138	6.2, 6.3 Data Links
REQ-083, REQ-118, REQ-122, REQ-125	6.4, 6.5 Reprogramming Links
REQ-094, REQ-095, REQ-115	6.6 Operator Interface

C Open Items

The values below are not yet known.

TBD-1 Project name and document reference.

TBD-2 Commands to the controlled equipment, the modes that accept them, and which are dangerous; the equipment is outside this project.

TBD-3 Time after which an arming command expires [3, REQ-030].

TBD-4 Largest allowed difference between the controller's clock and a scheduled run time [3, REQ-134].

TBD-5 Rate of regular health reports [3, REQ-037].

TBD-6 Interval of the link check on each data link.

TBD-7 Number of missed answers in a row after which a data link is reported lost.

TBD-8 CAN bit rate of data link B.

TBD-9 Number of tries of each reprogramming link before a restore moves on.

TBD-10 Electrical form of the SWD, NRST and BOOT lines between the units.

All statements in this document were stated or accepted by the author on 12 October 2026.